Region 5

Set up all six orders of integration for

$$\iiint_{R_5} f(x,y,z)\,dV,$$

where R_5 is the region bounded by

$$y+z=2,\ x=4-y^2,\ x=0,\ y=0,\ \text{and } z=0.$$

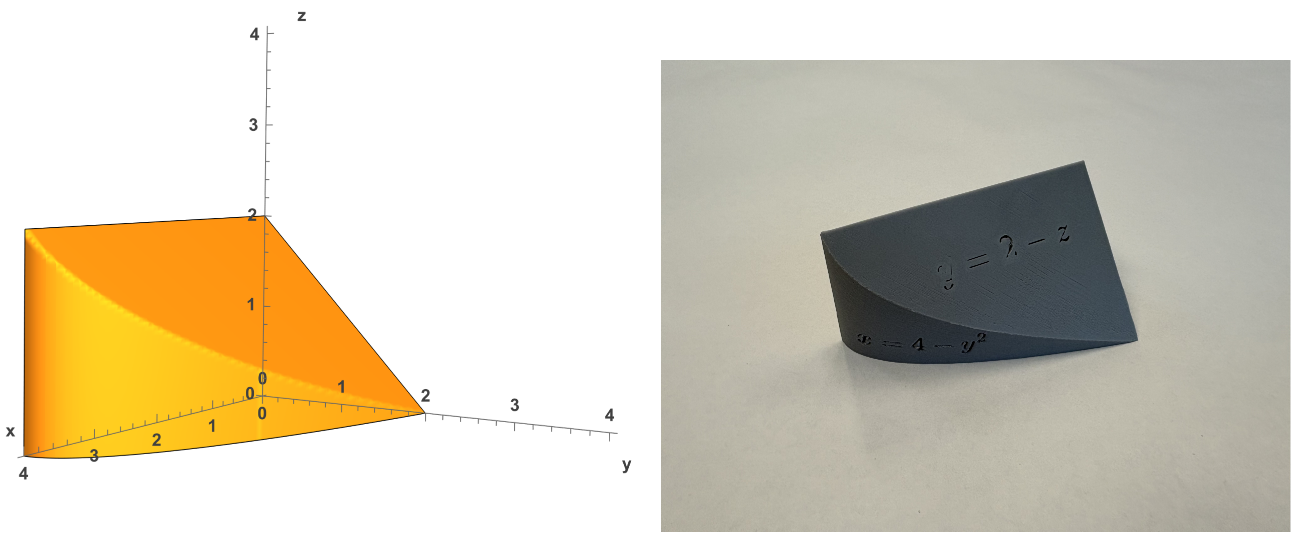
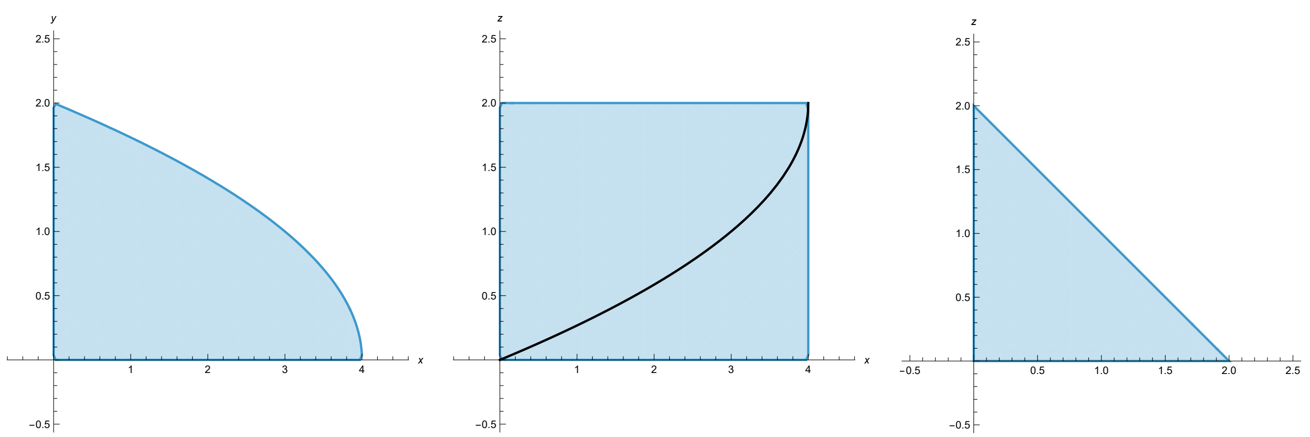


Figure 1. Computer-modeled and 3D-printed views of R_5



(A) xy-projection. (B) xz-projection. (C) yz-projection.

Figure 2. Projections of R_5 into the three coordinate planes.

The following triple integrals represent the solid using the xy-projection:

$$\int_0^2 \int_0^{4-y^2} \int_0^{2-y} f(x,y,z)\,dz\,dx\,dy$$
$$\int_0^4 \int_0^{\sqrt{4-x}} \int_0^{2-y} f(x,y,z)\,dz\,dy\,dx$$

The following triple integrals represent the solid using the xz-projection:

$$\int_0^2 \int_0^{-z^2+4z} \int_0^{2-z} f(x,y,z)\,dy\,dx\,dz + \int_0^2 \int_{-z^2+4z}^{4} \int_0^{\sqrt{4-x}} f(x,y,z)\,dy\,dx\,dz$$
$$\int_0^4 \int_0^{2-\sqrt{4-x}} \int_0^{\sqrt{4-x}} f(x,y,z)\,dy\,dz\,dx + \int_0^4 \int_{2-\sqrt{4-x}}^{2} \int_0^{2-z} f(x,y,z)\,dy\,dz\,dx$$

The following triple integrals represent the solid using the yz-projection:

$$\int_0^2 \int_0^{2-z} \int_0^{4-y^2} f(x,y,z)\,dx\,dy\,dz$$
$$\int_0^2 \int_0^{2-y} \int_0^{4-y^2} f(x,y,z)\,dx\,dz\,dy$$

All of these triple integrals evaluate to $\frac{20}{3}$ when $f(x,y,z) = 1$.